Script-Controlled ACL: Restrict Record Access Based on Field Value

Ideation Phase: Empathy Map Canvas

Date: October 3, 2026	**Maximum Marks:** 4 Marks
Team ID: SWTID-2026-8089	**Project Name:** Script-Controlled ACL

Empathy Map Canvas: Overview

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviors and attitudes. It is a useful tool to help teams better understand their users. Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise helps participants consider things from the user's perspective along with his or her goals and challenges.

- **Target Persona Focus:** ServiceNow Administrator / Security Administrator responsible for protecting student records.

Empathy Map Quadrants (Say, Think, Do, Feel)

The empathy map is organized into four core quadrants surrounding the user:

1. Says (verbatim quotes or common expressions from users)

- "Why can every user see these student records?"
- "I need each role to do only its own job: read, create, edit or delete."
- "EEE branch records must be visible only to EEE users, while admins keep full access."

2. Thinks (underlying beliefs, thoughts and assumptions)

- **Beliefs:** Default table access is too open and cannot protect sensitive data on its own.
- **Concerns:** Worrying that unauthorized users may view, change or delete student records.
- **Aspirations:** Wishing for automatic, role-based and field-based access rules that need no manual checking.

3. Does (observable actions and behaviors)

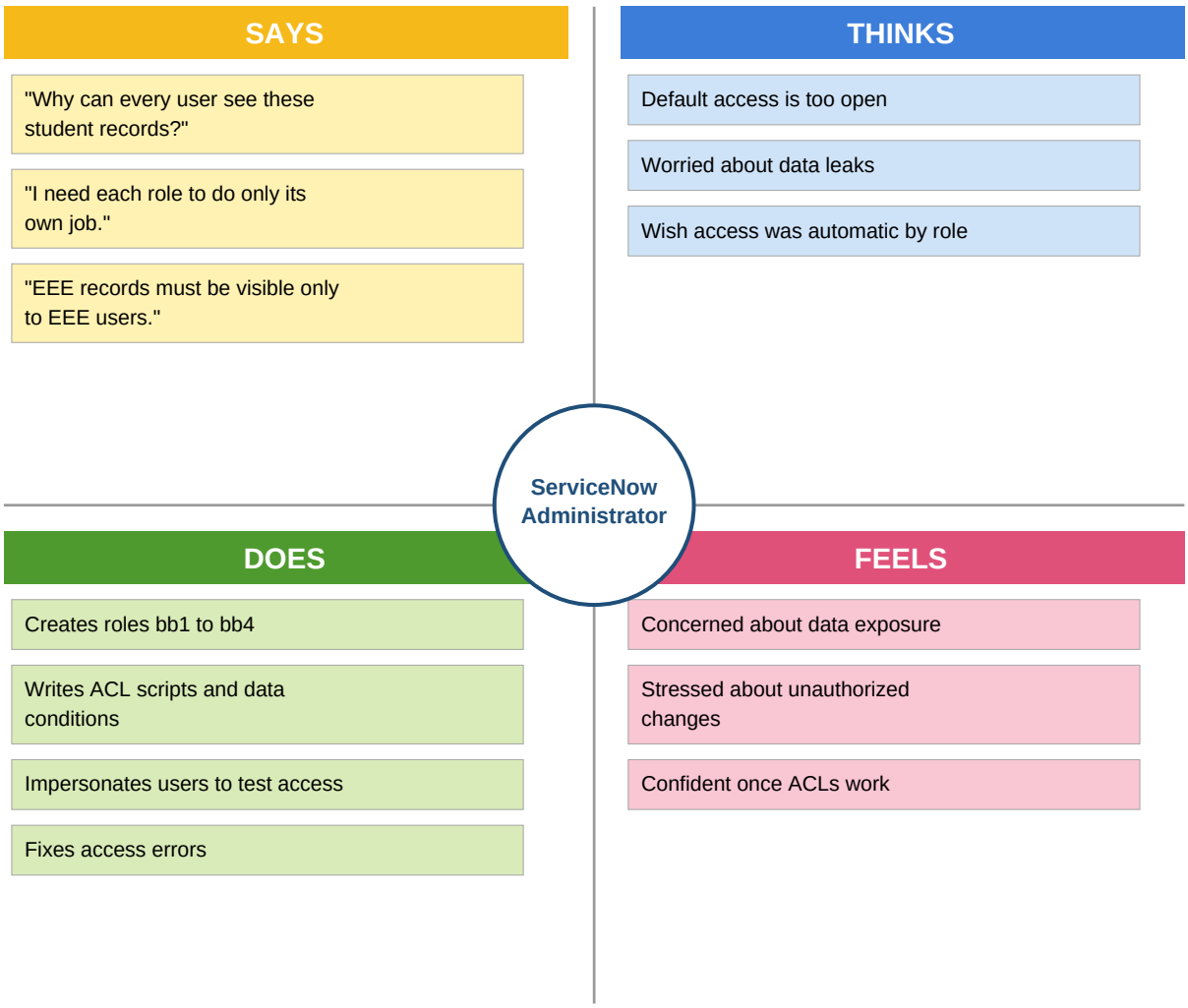
- Creates the EEE test user and the custom roles bb1, bb2, bb3 and bb4.
- Creates the Institution Details table with the Branch field (ECE, EEE, CSE) and adds sample records.
- Elevates to security_admin and writes READ, CREATE, WRITE and DELETE ACLs with a data condition and script.
- Impersonates the EEE user, a user without roles and the admin to verify each result.

4. Feels (emotions, frustrations and anxieties)

- **Concerned:** Anxious about sensitive student data being exposed to the wrong users.
- **Stressed:** Worried about accidental edits or deletions by users who should not have that permission.
- **Confident / Relieved:** Reassured when users with bb1 see only EEE records, users without roles see nothing, and admins see everything.

Pains and Gains Summary (Optional Extension)

- **Pains:** Open access to sensitive records, no separation between read, create, edit and delete permissions, risk of data leaks, and time spent testing access manually.
- **Gains:** Record-level security based on role and Branch value, clear separation of permissions through bb1 to bb4, full admin control, and protected student data.



ACL configuration used to address these needs:

servicenow

Access Control: u_institution_details

Type	record
Operation	read
Name	Institution Details [u_institution_details]
Data Condition	Branch is EEE
Requires role	bb1
Advanced script	gs.hasRole('bb1') returns true